PLCGen Custom Question Bank

Custom Question Bank dataset section

Contents

1 Custom Question Bank **2**
1.1 Overview and Naming 2
1.2 Source and Adaptation Record 2
1.3 Rationality of the Scenarios 2
1.4 Diversity Measurement 4
1.5 Formal Specifications 4
1.6 Test Cases and Validation 4

A Custom Question Bank LTL Specifications **5**

1 Custom Question Bank

1.1 Overview and Naming

The Custom Question Bank contains 12 original PLC programming tasks. They are named `C01`–`C12` and implemented as function blocks `FB_C01`–`FB_C12`. The public question bank uses `FB_Q01`–`FB_Q20`; the `C` prefix therefore keeps the custom blocks separate from the public ones.

The 12 tasks are deliberately balanced across three axes: sorting/filtering, cyclic timing and safety logic. The public bank is dominated by state machines and alarm/debounce behaviour, so these axes add complementary coverage. Each task is stored in `data/custom_questions.json` with its interface, difficulty, rationale, source record, LTL specification and test cases.

Table 1: Custom Question Bank at a glance. I/O counts include all function-block inputs and outputs.

ID	POU	Title	Difficulty	Axis	I/O	Props.	Tests
C01	`FB_C01`	Product Reject and Filtering Station	Easy	Sorting / filtering	4/5	3	2
C02	`FB_C02`	Two-Hand Safety Press	Easy	Safety logic	4/3	3	2
C03	`FB_C03`	Automatic Barrier Gate	Easy	Cyclic timing	3/3	3	2
C04	`FB_C04`	Star-Delta Motor Starter	Medium	Cyclic timing	4/5	3	2
C05	`FB_C05`	Car Wash Sequence	Medium	Cyclic timing	4/7	3	2
C06	`FB_C06`	Elevator Door Control	Medium	Safety logic	6/6	3	3
C07	`FB_C07`	Bottle Height Sorting	Medium	Sorting / filtering	4/6	3	2
C08	`FB_C08`	Greenhouse Irrigation Zones	Medium	Cyclic timing	6/5	3	3
C09	`FB_C09`	Conveyor Zone Accumulation	Hard	Sorting / filtering	6/7	3	3
C10	`FB_C10`	Boiler Purge and Ignition Sequence	Hard	Safety logic	5/6	3	3
C11	`FB_C11`	Palletizer Layer Counting	Hard	Sorting / filtering	3/5	3	1
C12	`FB_C12`	Safety Light Curtain Muting	Hard	Safety logic	5/4	3	3

1.2 Source and Adaptation Record

All tasks are original scenarios written for this project; no external question text was copied. The scenarios were synthesised from standard industrial PLC control patterns (IEC 61131-3 function blocks, safety interlocks, conveyor sequencing, batching and flame-proving sequences). The table below records the source type of each scenario.

The adaptation procedure is common to all tasks. Each scenario was manually reduced to a single function block with deterministic scan-cycle behaviour, bounded timers and no vendor-specific libraries. Each task received an internal review against PLC realism; independent peer review is still pending.

1.3 Rationality of the Scenarios

Each scenario reflects a recognisable PLC function. The list below records why the task is realistic and which control problem it exercises.

Table 2: Source record. All 12 scenarios use independent conception.

ID	Scenario	Source type
C01	Product Reject and Filtering Station	Independent conception
C02	Two-Hand Safety Press	Independent conception
C03	Automatic Barrier Gate	Independent conception
C04	Star-Delta Motor Starter	Independent conception
C05	Car Wash Sequence	Independent conception
C06	Elevator Door Control	Independent conception
C07	Bottle Height Sorting	Independent conception
C08	Greenhouse Irrigation Zones	Independent conception
C09	Conveyor Zone Accumulation	Independent conception
C10	Boiler Purge and Ignition Sequence	Independent conception
C11	Palletizer Layer Counting	Independent conception
C12	Safety Light Curtain Muting	Independent conception

C01 — Product Reject and Filtering Station A reject station is a common end-of-line filtering task: good and defective parts must be classified, counted and separated without stopping the conveyor until a batch is complete.

C02 — Two-Hand Safety Press Two-hand control prevents an operator from reaching into a press while it closes; the anti-tiedown timer catches a blocked or defeated button.

C03 — Automatic Barrier Gate Vehicle barriers combine a short open command, a hold-open condition and a delayed close; the safety beam must reopen the gate at any time.

C04 — Star-Delta Motor Starter Star-delta starting is a standard motor-control sequence that requires a timed transition, contactor interlocking and a latched overload fault.

C05 — Car Wash Sequence A car wash is a clean multi-step cyclic process: foam, brush, rinse and dry must run in order, and an emergency stop must abort the sequence safely.

C06 — Elevator Door Control Elevator doors combine limit switches, an obstacle detector and an automatic close timer; the obstacle must always win over a close command.

C07 — Bottle Height Sorting Height sorting is a realistic quality-gate task: an analog measurement must be filtered into accept/reject classes and the two streams counted independently.

C08 — Greenhouse Irrigation Zones Zone irrigation is a cyclic scheduling task with analog moisture thresholds; dry zones are watered in order and wet zones are skipped.

C09 — Conveyor Zone Accumulation Zone accumulation is a core material-handling function: parts must queue without collision, and a blocked zone must raise a jam alarm instead of stalling silently.

C10 — Boiler Purge and Ignition Sequence Burner start-up is safety-critical: a purge must precede ignition, flame must be proven within a fixed window, and failure must latch a lockout.

C11 — Palletizer Layer Counting Palletizing combines product counting, layer formation and a full-pallet handshake; it is a realistic batching and filtering problem.

C12 — Safety Light Curtain Muting Muting lets a pallet pass through a light curtain without tripping the safety stop, but only when two sensors confirm the correct sequence; a wrong sequence must alarm.

1.4 Diversity Measurement

Diversity is measured along two dimensions: the primary control axis and the difficulty level. Each primary axis contains four questions, and the difficulty distribution is 3 easy, 5 medium and 4 hard.

Table 3: Diversity by primary control axis.

Axis	Questions	Count
Sorting / filtering	C01, C07, C09, C11	4
Cyclic timing	C03, C04, C05, C08	4
Safety logic	C02, C06, C10, C12	4

Table 4: Difficulty distribution.

Difficulty	Questions	Count
Easy	C01, C02, C03	3
Medium	C04, C05, C06, C07, C08	5
Hard	C09, C10, C11, C12	4

Additional diversity comes from the required skills: edge detection, timers, counters, analog thresholds, state machines, safety interlocks, jam detection and flame proving. The complete per-question tags are stored in the dataset JSON.

1.5 Formal Specifications

Every question is specified in linear temporal logic (LTL). Each question has three properties: a safety or invariant property, a response or sequencing property, and a recovery or termination property. Numeric conditions are represented as derived atoms, and timer conditions are represented by internal atoms mapped to the reference implementation's timer output. The full set of 36 formulas is given in Appendix A.

The specifications assume a periodic scan (20 ms in the test cases) and that inputs change only at scan boundaries. The reference ST implementation is used only to validate the specifications and test cases; it is not shown to the coding agent.

1.6 Test Cases and Validation

Test cases follow the repository format used by `codesys_service/run_tests.py`. Each question has a JSON test-case file with interface declarations and one or more scenarios. A scenario contains inputs, expected outputs and optional hold times for timers. The cases cover initialisation, normal operation, priority or edge conditions, and fault recovery where applicable.

The local validation reports 28/28 simulation cases passing, and all 12 reference implementations pass the static check. The real CODESYS compilation results are reported by the experimental evaluation. The tests validate functional behaviour against the reference ST; they do not replace a CODESYS compile in the target environment.

Table 5: Test-case coverage and local validation results.

ID	Test cases	Steps	Dry check	Sim pass
C01	2	16	VALID	2/2
C02	2	10	VALID	2/2
C03	2	9	VALID	2/2
C04	2	9	VALID	2/2
C05	2	11	VALID	2/2
C06	3	10	VALID	3/3
C07	2	8	VALID	2/2
C08	3	7	VALID	3/3
C09	3	8	VALID	3/3
C10	3	13	VALID	3/3
C11	1	14	VALID	1/1
C12	3	13	VALID	3/3

A Custom Question Bank LTL Specifications

This appendix lists the three LTL properties for each custom question. Atoms, assumptions and machine-readable formulas are stored in `specs/custom/Cxx.ltl.json`. The operators are: **G** (always), **F** (eventually), **X** (next scan), $\wedge$ (and), $\vee$ (or), $\neg$ (not) and $\rightarrow$ (implies).

C01 — Product Reject and Filtering Station

POU: `FB_C01`; file: `specs/custom/C01.ltl.json`.

- **P1 — Defective part is rejected**: $\mathbf{G}((PartPresent \wedge DefectSensor \wedge RunSwitch \wedge \neg BatchFull) \rightarrow RejectPusher)$
- **P2 — Pusher retracts**: $\mathbf{G}(RejectPusher \rightarrow \mathbf{F}\neg RejectPusher)$
- **P3 — Batch full stops the conveyor**: $\mathbf{G}(BatchFull \rightarrow \neg ConveyorMotor)$

C02 — Two-Hand Safety Press

POU: `FB_C02`; file: `specs/custom/C02.ltl.json`.

- **P1 — Two-hand start**: $\mathbf{G}((LeftButton \wedge RightButton \wedge PartPresent \wedge \neg Fault) \rightarrow PressOutput)$
- **P2 — Part interlock**: $\mathbf{G}(PressOutput \rightarrow PartPresent)$
- **P3 — Anti-tiedown detects a held button**: $\mathbf{G}((LeftOnlyTimerQ \vee RightOnlyTimerQ) \rightarrow \mathbf{F}Fault)$

C03 — Automatic Barrier Gate

POU: `FB_C03`; file: `specs/custom/C03.ltl.json`.

- **P1 — Vehicle opens the gate**: $\mathbf{G}(VehiclePresent \rightarrow GateOpen)$
- **P2 — Beam opens and warns**: $\mathbf{G}(SafetyBeam \rightarrow (GateOpen \wedge WarningLight))$
- **P3 — Delayed close**: $\mathbf{G}((GateOpen \wedge \neg VehiclePresent \wedge \neg SafetyBeam) \rightarrow \mathbf{F}GateClosed)$

C04 — Star-Delta Motor Starter

POU: `FB_C04`; file: `specs/custom/C04.ltl.json`.

- **P1 — Overload trips all contactors**: $\mathbf{G}(Overload \rightarrow (\neg MainContactor \wedge \neg StarContactor \wedge \neg DeltaContactor))$
- **P2 — Star-delta interlock**: $\mathbf{G}(\neg(StarContactor \wedge DeltaContactor))$
- **P3 — Timed transition**: $\mathbf{G}((MainContactor \wedge StarContactor) \rightarrow \mathbf{F}DeltaContactor)$

C05 — Car Wash Sequence

POU: `FB_C05`; file: `specs/custom/C05.ltl.json`.

- **P1 — Emergency stop removes all actuators**: $\mathbf{G}(EmergencyStop \rightarrow \neg(ConveyorMotor \vee FoamValve \vee BrushMotor \vee RinseValve \vee DryFan))$
- **P2 — Foam requires conveyor**: $\mathbf{G}(FoamValve \rightarrow ConveyorMotor)$
- **P3 — Cycle eventually completes**: $\mathbf{G}((StartButton \wedge VehiclePresent \wedge \neg Fault) \rightarrow \mathbf{F}CycleComplete)$

C06 — Elevator Door Control

POU: `FB_C06`; file: `specs/custom/C06.ltl.json`.

- **P1 — Obstacle prevents closing**: $\mathbf{G}(ObstacleSensor \rightarrow \neg DoorMotorClose)$
- **P2 — Motor interlock**: $\mathbf{G}(\neg(DoorMotorOpen \wedge DoorMotorClose))$
- **P3 — Automatic close**: $\mathbf{G}((DoorOpenLimit \wedge \neg ObstacleSensor \wedge \neg OpenButton) \rightarrow \mathbf{F}DoorMotorClose)$

C07 — Bottle Height Sorting

POU: `FB_C07`; file: `specs/custom/C07.ltl.json`.

- **P1 — In-range bottle is accepted**: $\mathbf{G}((PartPresent \wedge HeightInRange \wedge RunSwitch \wedge \neg Fault) \rightarrow AcceptGate)$
- **P2 — Out-of-range bottle is rejected**: $\mathbf{G}((PartPresent \wedge HeightOutOfRange \wedge RunSwitch \wedge \neg Fault) \rightarrow RejectGate)$
- **P3 — Gates are mutually exclusive**: $\mathbf{G}(\neg(AcceptGate \wedge RejectGate))$

C08 — Greenhouse Irrigation Zones

POU: `FB_C08`; file: `specs/custom/C08.ltl.json`.

- **P1 — Valve implies pump**: $\mathbf{G}((Zone1Valve \vee Zone2Valve \vee Zone3Valve) \rightarrow Pump)$
- **P2 — Stop removes all outputs**: $\mathbf{G}(StopButton \rightarrow \neg(Zone1Valve \vee Zone2Valve \vee Zone3Valve \vee Pump))$
- **P3 — Started cycle terminates**: $\mathbf{G}((StartButton \wedge \neg CycleComplete) \rightarrow \mathbf{F}CycleComplete)$

C09 — Conveyor Zone Accumulation

POU: `FB_C09`; file: `specs/custom/C09.ltl.json`.

- **P1 — Zone 1 moves into a free zone 2**: $\mathbf{G}((Zone1Sensor \wedge \neg Zone2Sensor \wedge RunSwitch) \rightarrow Zone1Motor)$
- **P2 — Zone 1 full condition**: $\mathbf{G}((Zone1Sensor \wedge Zone2Sensor) \rightarrow Zone1Full)$
- **P3 — Jam stops all motors**: $\mathbf{G}(JamAlarm \rightarrow \neg(Zone1Motor \vee Zone2Motor \vee Zone3Motor))$

C10 — Boiler Purge and Ignition Sequence

POU: `FB_C10`; file: `specs/custom/C10.ltl.json`.

- **P1 — Lockout removes all combustion outputs**: $\mathbf{G}(Lockout \rightarrow \neg(PurgeFan \vee Igniter \vee FuelValve))$
- **P2 — Fuel requires proven flame**: $\mathbf{G}(FuelValve \rightarrow FlameOn)$
- **P3 — Failed ignition locks out**: $\mathbf{G}((Igniter \wedge \neg FlameSensor) \rightarrow \mathbf{F}Lockout)$

C11 — Palletizer Layer Counting

POU: `FB_C11`; file: `specs/custom/C11.ltl.json`.

- **P1 — Layer completion pushes**: $\mathbf{G}(LayerComplete \rightarrow LayerPush)$
- **P2 — Full pallet stops conveyor**: $\mathbf{G}(PalletFull \rightarrow \neg ConveyorMotor)$
- **P3 — Push pulse ends**: $\mathbf{G}((LayerPush \wedge \neg PalletFull) \rightarrow \mathbf{F}\neg LayerPush)$

C12 — Safety Light Curtain Muting

POU: `FB_C12`; file: `specs/custom/C12.ltl.json`.

- **P1 — Curtain stops the conveyor**: $\mathbf{G}((LightCurtain \wedge \neg MutingActive) \rightarrow SafetyStop)$
- **P2 — Safety stop stops the conveyor**: $\mathbf{G}(SafetyStop \rightarrow \neg ConveyorMotor)$
- **P3 — Valid sequence activates muting**: $\mathbf{G}((MutingSensor1 \wedge \mathbf{X}MutingSensor2) \rightarrow \mathbf{F}MutingActive)$